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RESEARCH ARTICLE

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Elucidating the impacts of trees on landslide initiation throughout a typhoon: Preferential infiltration, wind load and root reinforcement

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Abstract

Typhoon-induced landslides are widespread, damaging and deadly. In this study, we elucidated their spatiotemporal features through investigations in Southeast coastal China and proposed a numerical model that involves typhoon wind load, preferential infiltration and root reinforcement. The wind load is calculated through a combination of the autoregressive method and a multi-degree-of-freedom tree swaying mode, transmitting to the slope through the wind–tree interaction. Taproots of trees are modelled as piles, and root–soil interfaces are modelled as preferential infiltration boundaries. Using the numerical model, we quantitatively assessed the impacts of wind load, rainfall infiltration and root reinforcement on the slope stability and compared results with cases in Waipaoa and Wairoa (New Zealand). Results suggest that the impacts of trees depend heavily on meteorological and geological conditions. Trees play a destabilizing role in Southeast China during a typhoon but a stabilizing role in the cases in New Zealand. For slopes in Southeast China with a thick soil layer (>root depth), strong wind load and preferential flow resulting from root–soil interfaces, rather than slope surface infiltration, significantly decrease the slope stability in a short time. Slope failure occurred in all scenarios that account for the preferential infiltration, and its combination with a Force 14 typhoon wind load can fail the slope at 10 h after the typhoon initiation. Differently, tree roots in cases in New Zealand can penetrate through the thin soil layer (<1 m) and provide considerable additional resistance to stabilize the slope.

KEYWORDS

impact of trees, preferential infiltration, stability analysis, typhoon-induced landslide, wind load

1 | INTRODUCTION

Typhoon events are often linked to atmosphere variation and, therefore, are expected to increase in response to climate change (Elsner et al., 2008; Webster et al., 2005). According to the statistics, a total of 493 typhoons landed in China from 1949 to 2018, making China one of the countries suffering the most severe typhoon damage in the world (Cui et al., 2022; Jiao, 2021). These typhoons generated many landslides in the struck area and led to an enormous toll on fatalities and property. For example: In 2019, Typhoon Lekima triggered 251 shallow landslides and 82 debris flows in Daoshi Town, Zhejiang, China, causing 7 deaths and economic loss of 1.5 hundred million dollars (Liang et al., 2022). Interestingly, typhoon-induced landslides in Southeast China primarily occur on trees-covered slopes before and within 12 h after the typhoon landing, at approximately the same time as typhoon strikes (Zhuang et al., 2022). They show a weak time-lag effect and special spatial distribution (clustered on trees-covered slopes) and thus indicate their special failure mechanisms and disaster-causing factors.

Previous studies of typhoon-induced landslides have investigated the impact of rainfall on the slope stability, including the rainfall intensity and duration (e.g., Dou et al., 2019; Liang et al., 2022; Saito et al., 2014; Shou & Lin, 2020; Tsou et al., 2011; Zhang et al., 2019). They treated typhoon-induced landslides as rainfall-induced landslides and mainly analysed the impact of sub-surface hydrology on the slope stability. Fundamental questions have remained, however, about the impacts of vegetation and typhoons on the slope failure. Strong wind load can act on trees and transmit to the slope to work as a driving force (Buma & Johnson, 2015; Parra et al., 2021), while trees themselves play a complex role in slope stability (Pawlik, 2013).

Vegetation has been widely used for stabilizing slopes because of root reinforcement and beneficial hydrological effects (canopy interception and plant water transpiration) (Gonzalez-Ollauri & Mickovski, 2017; Hales, 2018; Ni et al., 2017). However, there are substantial doubts about how vegetation behaves in a typhoon event. The extra resistance provided by roots disappears below the root-reinforced area (Schwarz et al., 2010), and the beneficial hydrological effects potentially disappear in a rainstorm (McGuire et al., 2016). Therefore, these contributions hardly work below the rooting zone during a typhoon. Studies of vegetation impact on the slope stability have highlighted the potential for some destabilizing effects. Root–soil interfaces and decayed root channels create preferential flow paths that increase the soil infiltration rate (Ghestem et al., 2011; Jiang et al., 2018; Qin et al., 2022). Compared with a uniform infiltration, the detrimental hydrological effect significantly accelerates the water infiltration and provides conditions for rapid landslide initiation. Furthermore, Zhuang et al. (2022) suggested that typhoon-associated wind load is influential on the slope failure, typically for slopes covered by large trees. The preferential hydrological effect, combined with strong wind load and root reinforcement, necessitates a careful assessment of trees impact on the slope stability in a typhoon event. To our knowledge, no targeted research that jointly involves the above detrimental preferential hydrological effect, wind load and root reinforcement has been performed to quantitatively analyse the impact of trees on typhoon-induced slope failures to date.

Here, we first investigated 303 typhoon-induced landslides in Southeast coastal China and briefly summarized their general features. A targeted numerical model was then established for the failure mechanism analysis of typhoon-induced landslides, incorporating the impacts of wind, rainfall and trees. Specifically, the impacts of wind load (transmit to the slope from wind–tree interaction), preferential flow from root–soil interface and root reinforcement are involved in the model. This study aims to illustrate the impacts of trees on landslide initiation in Southeast coastal China throughout a typhoon and elucidate the possible failure mechanisms of typhoon-induced landslides. Our study fills the research gap in typhoon-induced landslides and provides new insights into further research.

2 | TYPHOON-INDUCED LANDSLIDES

Field investigations are focused on two provinces in Southeast coastal China: Zhejiang and Fujian (Figure 1a) (Zhuang et al., 2022).

Typhoon-induced landslides are clustered on trees-covered slopes. Among the 303 identified landslides, 68% of events are shallow soil landslides with a depth of <3 m and 73% of the identified events show a volume of <500 m^3 and an elevation difference of <10 m; 70% of events show a slope angle of 20–40° and reach a peak frequency of ∼30°. Landslide cases primarily occurred in the area of planar and convex upward topography; only 11% of cases occurred in areas of concave topography. Moreover, 283 of the total 303 investigated cases involve a cut slope (Figure 1e–g). Slopes in the area are characterized by a two-layer structure (upper layer of weathering-formed residual soil and lower layer of bed-rock). Sliding surfaces are concentrated on the rock–soil interface or the possible weak layer in the overburden soil. The spatial distribution of investigated events shows high agreement with the impact region of typhoons. Most landslides are identified within the radii of 7-scale and even 10-scale wind circles and the region with high rainfall intensity. For instance, landslides triggered by typhoons Morakot (2009) and Cimaron (2013) concentrated in zones within the Beaufort scale 7 wind circle and a cumulative rainfall greater than 200 mm (Figure 1b,c). Typhoon-induced landslides indicate a special temporal distribution, named 'weak time-lag effect', as events happen at about the same time as typhoon strikes; 87% of identified landslides occurred before and within 12 h after the typhoon landing and were in the peak period of wind force and rainfall intensity, showing significant differences with landslides triggered only by slope surface infiltration. Furthermore, all the investigated landslides occurred on slopes with high vegetation coverage (Figure 1d). Forest coverage (including trees, bamboo and some shrubs) in the study area reaches ∼60% (according to information provided by the Chinese government, Zhang, Liao, et al., 2022); 78% of the identified cases occurred on slopes covered by trees and bamboo. Pines (*Pinus massoniana*), which are most universally distributed in Southeast coastal China, show a mean height of 15–25 m and a distance of 2–5 m.

3 | METHODS AND DATA

3.1 | General model

To investigate the possible failure mechanism of typhoon-induced landslides coupled to wind, rainfall and trees in Southeast China, we established a targeted and novel numerical model using the commercial finite element method (FEM) ABAQUS. The two-dimensional (2D) generalized model of a cut slope was created with trees evenly distributed (Ng et al., 2016) in the shallow soil layer (Figure 2). The 2D FEM model has been widely used to analyse the slope stability (Griffiths et al., 2009; Ishii et al., 2012; Zheng et al., 2008), and the model geometry here is generalized based on *in situ* measurements. The model is a two-layer slope with a height of 20 m and a length of 50 m, and the initial underground water table was in the lower bed-rock. A slope angle of 30° and an overburdened soil layer of 3 m were selected for the generalized model because most investigated landslides have a slope of 20–40° and a depth of <3 m (parts sliding surface developed at the soil–rock interface). A cut slope is modelled in this study considering that 93% of the identified landslide cases occurred in a cut slope. Trees in the study area show a distance of 2–5 m, and here, a distance of 3 m is selected for the model.

In this study, pines (*P. massoniana*) commonly distributed in Southeast coastal China were selected for the modelling, and the corresponding parameters are presented in Table 1. This tree species has a strong taproot system but sparse lateral roots, and the taproot can penetrate to a depth of 2 m (Wang, 2014; Yang et al., 2021; Yao & Wang, 2020; Zhang, Yang, et al., 2022). Liang et al. (2015, 2020) conducted numerical work and laboratory tests and showed great performance in simulating taproots roots as piles. Wu (2013)

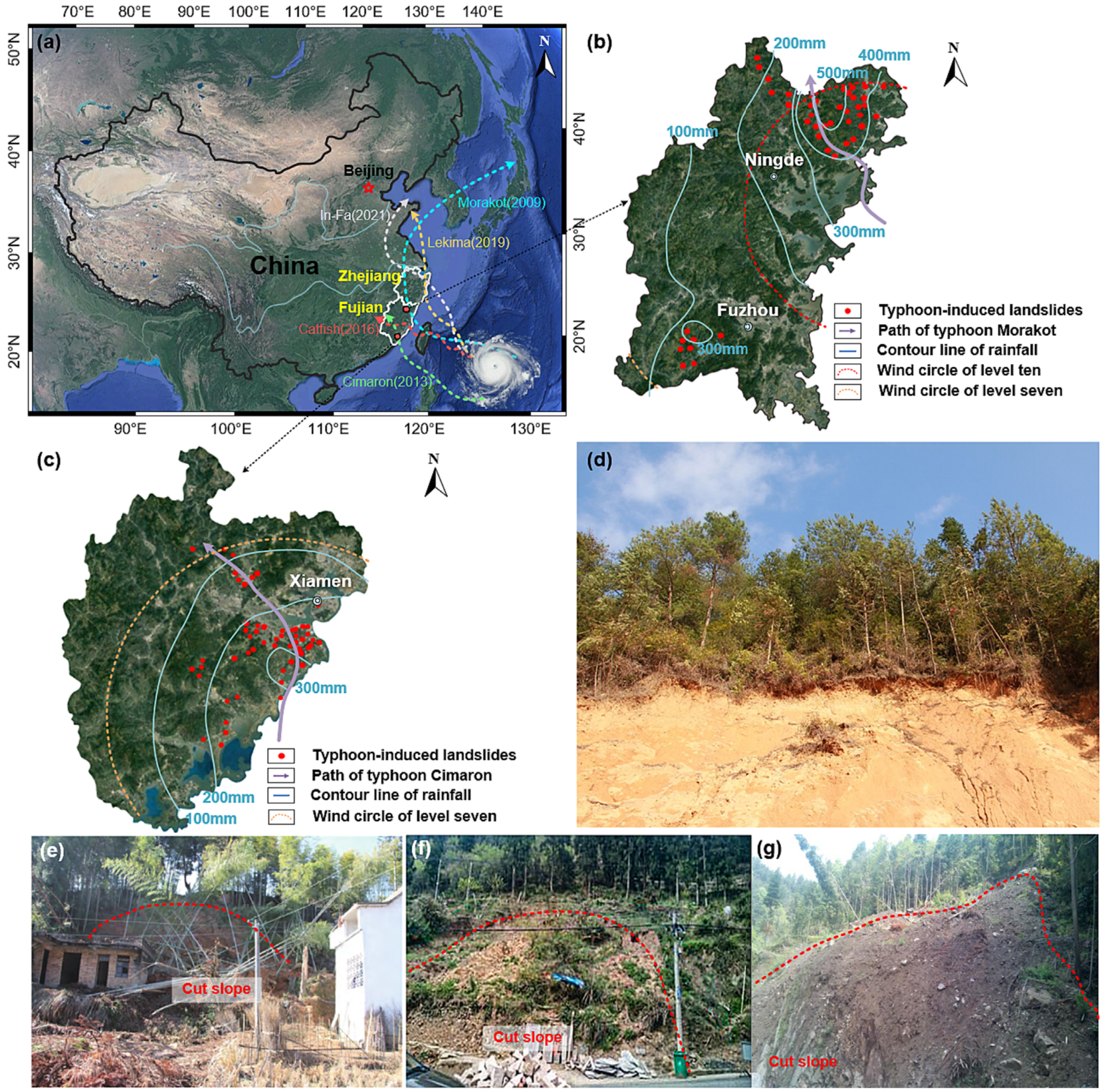

FIGURE 1 (a) Typical catastrophic typhoons that struck Southeast coastal China. (b, c) Spatial distribution of landslides triggered by the typhoons Morakot and Cimaron in Fujian. Wind circle represents the power and impact area of a typhoon (Geng et al., 2018; Li, Lin, et al., 2023), and the scale refers to the wind force listed in the Beaufort wind scale. (d) A typhoon-induced slope failure developed in the trees-covered slope, showing the heavily forested hillslope. (e–g) Landslide cases occurred on cut slopes that are close to the roads and houses. [Color figure can be viewed at wileyonlinelibrary.com]

and Dyson et al. (2023) also supported that root structures of primary taproots can be directly modelled using beams and piles. Therefore, to simply model the widely distributed trees that have a deep taproot in the area, each root was separated as a pile embedding in the soil layer and contact pairs are defined at root–soil interfaces. Normal contact obeys 'hard contact' (Khalili et al., 2011), indicating that the contact constraint is applied when the clearance between two surfaces is zero and the contact surfaces will separate when contact pressure is equal to 0. The tangential contact behaviour is described by the friction coefficient between the pile and soil and is taken as $\mu = tan(\varphi)$ (φ is the soil friction angle) (Rizvi et al., 2022). Roots are modelled with a decreasing diameter from the ground surface to the maximum depth, which play the role to reinforce the soil and transmit the wind load to the slope.

Furthermore, for the soil and rock parts, the Mohr–Coulomb criterion (Vanapalli et al., 1996) was used to describe their shear strength behaviours, and their permeability properties are represented by the Van Genuchten (1980) model. According to our measurements of historical cases in the study area (Zhuang et al., 2022), soil deposits in the study area show an effective frictional angle φ of 16–25°, an effective cohesion c of 2–8 kPa and an average saturated permeability coefficient of $\sim 4 \times 10^{-6}$ m/s. Therefore, median quartile values of the given scope are selected for the modelling (frictional angle of 20.5° and effective cohesion of 5 kPa). The bottom boundary of the model is fully fixed in both horizontal and vertical directions, and the lateral boundaries are fixed in the horizontal direction. A total of 4849 meshes and 5253 nodes are generated for the entire model (Figure 3). Input parameters for the numerical model are presented in Table 2.

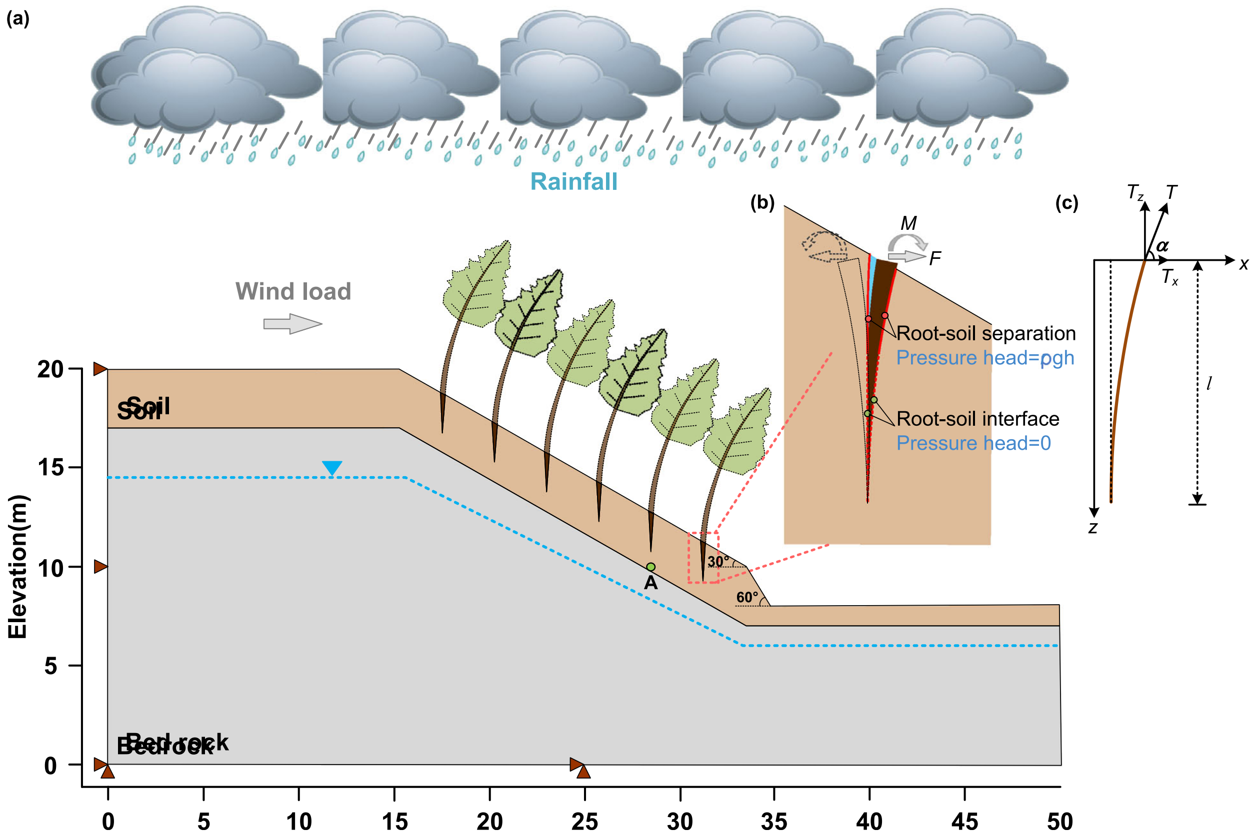

FIGURE 2 (a) Schematic model of the typhoon-induced landslide. 'A' is the monitoring point. (b) Principle of the preferential infiltration boundary. A head boundary increasing with depth is set in the separated region, and a zero-head boundary is assumed in the lower region. (c) Validity of modelling taproots as piles using Equation (5) (Wu, 2013). The brown line represents the root. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Model parameters used in the numerical simulations of the tree response (according to Zhuang et al., 2022).

Parameters for the modelling of the tree response	Unit	Value
Surface roughness coefficient α	-	0.3
Order of AR model p	-	4
Tree trunk mass	kg	350
Tree crown mass	kg	55
Tree height h_t	m	20
Crown width	m	5
Crown height	m	10
Diameter at the base of the tree trunk	m	0.25
Tree distance L	m	3
Root depth	m	2
First modal mass m	kg	11.2
First modal damping c	kg/s	10.6
First modal stiffness k	kg/s^2	62.8
Vibration frequency of the first mode	Hz	0.372
Drag coefficient C_d	-	0.35
Length of beam segment ds	m	1

Abbreviation: AR, autoregressive.

3.2 | Wind load modelling

The modelling procedure consists of two stages. First, the model is run under gravity and an initial underground water table to reach the initial stresses. Second, the core of the model, applying the wind and hydrological load to investigate the fluid pressure, plastic strain and deformation of the slope. Specifically, the wind load transmitted to the slope is modelled through combining the autoregressive method and a modified multi-degree-of-freedom tree swaying model. The autoregressive method accounts for the fluctuation component of the wind velocity and ground roughness and has been widely used to calculate the wind speed time series (Jeong & Bienkiewicz, 1997; Poggi et al., 2003; Santamaría-Bonfil et al., 2016). The modified multi-degree-of-freedom tree swaying model treats the tree as a flexible cantilever beam that is clamped in the ground (Pivato et al., 2014; Zhuang et al., 2022). The tree model accounts for the inclination and eccentric gravity of trees and calculates the tree deformation through a linear modal analysis. The method is well described in Zhuang et al. (2022) and is not introduced in detail here. The wind modelling equations are shown below:

$$u(t) = \bar{u} + u'(t) = \bar{u} + \sum_{j=1}^{p=4} \varphi_j u'(t - j\Delta t) + N(t) \quad (1)$$

$$m\frac{\partial^2 y}{\partial t^2} + c\frac{\partial y}{\partial t} + ky = \int_0^h F_{wi}\varphi ds + \int_0^h m_i g \cdot sin\theta_i \cdot cos\theta_i \varphi ds \quad (2)$$

$$F_{wi} = \rho C_d A_f \left| ucos\theta_i - \frac{\partial y}{\partial t}cos\theta_i \right| \left(ucos\theta_i - \frac{\partial y}{\partial t}cos\theta_i \right) cos\theta_i \quad (3)$$

where u is the wind velocity composed of a mean component $\bar{u}$ and fluctuating component u'; φ is the autoregressive parameter; N represents a random matrix; m, c and k represent the first modal mass,

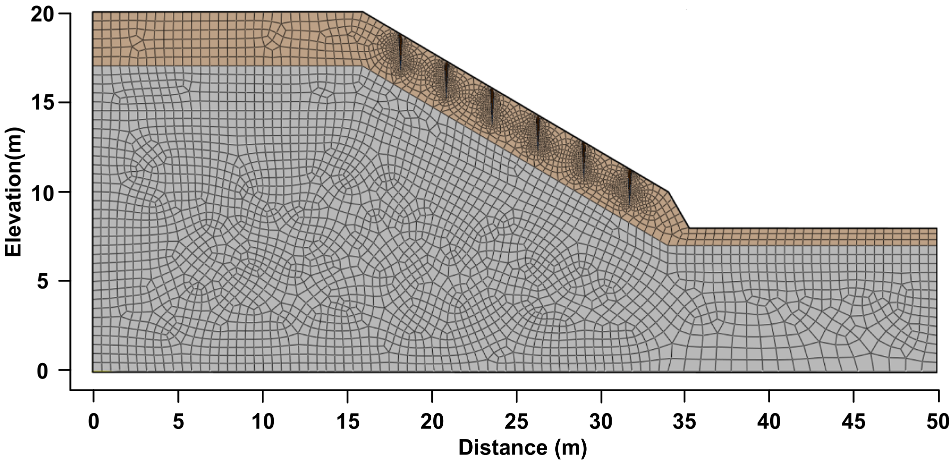

FIGURE 3 Mesh of the numerical model. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 2 Model parameters used in the numerical simulations of slope stability.

Material	Density (kg/m^3)	Elastic modulus (MPa)	Poisson's ratio	Effective cohesion (kPa)	Friction angle (°)	Saturated permeability coefficient ($\times$ 10^{-7} m/s)	Van Genuchten model α	Van Genuchten model *n*
Soil	1900	16	0.31	5	20.5	40	0.044	2.9
Bed-rock	2200	5080	0.245	200	36	1	0.044	2.9
Root	550	8100	0.41	-	-	-	-	-

Note: Parameters for the root are inferred from Gao et al. (2016).

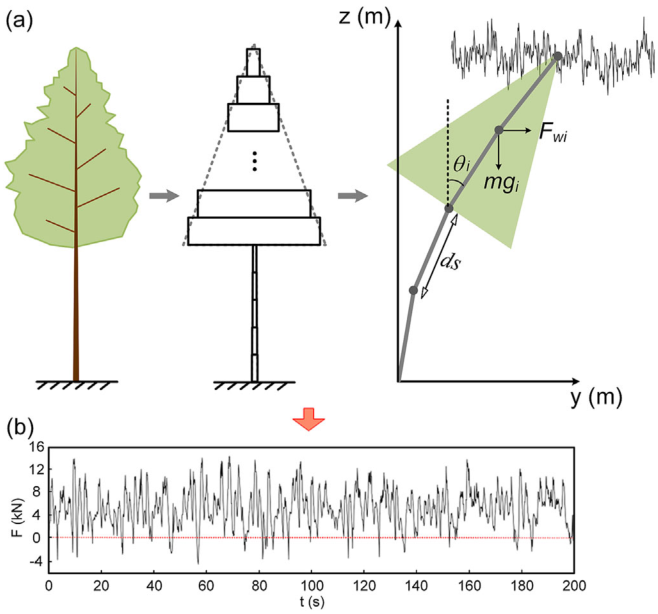

FIGURE 4 (a) Modelling tree as a multi-degree-of-freedom beam with variable cross-sections to simulate the wind force transmitted to the slope. (b) Total force–time series at the tree base under a Force 14 typhoon (positive and negative values mean that the force transmitted to the slope is the same or opposite to the wind direction). [Color figure can be viewed at wileyonlinelibrary.com]

damping and stiffness; φ is the first mode shape; ρ is the air density; C_d is the drag coefficient; A_f is the frontal area; θ represents the beam inclination; F_{wi} is the drag force; and $m_i g \cdot sin\theta_i \cdot cos\theta_i$ represents the contribution of eccentric beam gravity.

This method accounts for large tree deflections and the inertia force results from the tree motion (Figure 4), showing good performance in modelling the mechanical response of trees in a typhoon. The wind is set to blow in the downhill and uphill directions to simulate the most extreme scenario. The maximum value of the calculated force time series (at the tree base) was applied at the top of each root to simulate the maximum wind force of gust wind transmitted to the slope (Figure 4b). The windthrow was not accounted for because we did not observe the extensive occurrence of windthrow in a specific area.

3.3 | Preferential infiltration modelling

Hydrological load modelling is integrated as a uniform component and a preferential component. A combined flux–head boundary was set at the slope surface to simulate uniform infiltration. For an FEM mode, Cai and Ugai (2004) and Tian et al. (2017) suggest a flux boundary if all rainfall can penetrate into the slope and a zero-pressure head boundary when rainfall cannot completely infiltrate into the slope. Here, the flux boundary is changed to a head boundary when the rainfall intensity is larger than the saturated permeability of soil (hereinafter named intense rainfall) because surface run-off is generated and a zero-head boundary is suitable in such a scenario (Collins & Znidarcic, 2004). Furthermore, root–soil interfaces are suggested as the boundaries of preferential infiltration (Zhang et al., 2017). Ghestem et al. (2011), Zhang et al. (2017) and Li, Cui, and Yin (2023) performed laboratory tests and *in situ* dye disappearance tests, demonstrating that root–soil gaps have a width of millimetres and can significantly accelerate water infiltration. According to Ghestem et al. (2011) and our preliminary wind tunnel tests (Sun et al., 2021), the violent tree swaying in a typhoon can intensify this hydrological effect because the upper part of taproots will separate from the soil and results in a large crack. The regional soil permeability (near roots) can be several times the original state. To model this special hydrological loading, user subroutines DISP and URDFIL (Fortran-based codes) helped to model the boundary. The contact state at the root–soil interface is judged using the URDFIL subroutine at each time step (maximum time step of 10 s). A head boundary increasing with depth was set in the separated region to simulate the impact of large cracks (Davidson, 1985; Krisnanto et al., 2016; Yanamandra & Oh, 2021), and a zero-head boundary was assumed in the lower region using the DISP subroutine, regarding that contacted region of root–soil interface can hardly form a penetrating vertical crack (Figure 2b).

3.4 | Factor of safety calculation

The widely used 'shear strength reduction method' is employed to calculate the factor of safety (F_s) of the slope (Dawson et al., 1999; Gao et al., 2021):

$$F_s = \frac{c}{c_{cri}} = \frac{tan(\varphi)}{tan(\varphi_{cri})} \tag{4}$$

where c_{cri} and φ_{cri} are the artificially reduced shear strength parameters until the slope failure occurs.

3.5 | Model validation

The work checked the validity of modelling taproots as piles. Wu (2013) defined an index (ηl) to describe the root behaviour (Figure 2c):

$$\eta l = \sqrt{\frac{T_x \cdot tan\alpha}{EI}} \cdot l \tag{5}$$

where l is the root length; E is the elastic modulus; $I = \frac{\pi d^4}{64}$ is the moment of inertia; d is the diameter at the root top (tree base); α represents the inclination of a taproot relative to the horizontal direction; and T_x represents the horizontal force subjected by the root top, corresponding to the wind force in this study. The root reinforcement can be modelled as a pile subjected to lateral loads when $\eta l < 1.5$. Using tree-related parameters presented in Table 1, the maximum T_x was calculated to be 14.5 kN (Figure 4b) when the tree was subjected to a Force 14 typhoon. Equation (5) yields an ηl of <0.35 for the parameters of $l = 2$ m, $E = 8.1$ GPa and $d = 0.25$ m, significantly smaller than the threshold proposed by Wu (2013). Therefore, we suggested that modelling roots as piles is an applicable approach in this study.

To investigate the impacts of different factors on the slope stability, the work further checked the validity of the numerical model. For the calculation of F_s of a slope, Cheng et al. (2007) and Liu et al. (2015) suggest a high agreement between the calculated F_s from the 2D FEM method and the limited equilibrium method (LEM). Therefore, the model is checked through comparing the modelling results with the LEM presented in Zhuang et al. (2022). We calculated cases of wind blowing downhill, and the error is defined as $\frac{Fs(FEM) - Fs(LEM)}{Fs(FEM)}$, as presented in Figure 5. Results indicated that though the F_s calculated from the LEM is slightly lower than the FEM, the calculated results are in good agreement. The error shows a range of 1.0%–4.7%; thus, we suggest that the proposed model is applicable for the following analysis.

In this study, rainfall data of Typhoon Lekima (2019) recorded by the Yongjia meteorological bureau in Zhejiang (Figure 6) and Force 3–14 winds (mean velocity of 3–45 m/s) on the Beaufort scale (Table 3) that cover almost all the forces in Southeast coastal China were utilized for the numerical model.

4 | RESULTS

In a typhoon event, the areas with high-intensity rainfall can coincide with the area experiencing strong wind; therefore, it is hard to separate the wind effects from the effects of high-intensity rainfall (Figure 1b,c). However, the characteristics of typhoon-induced landslides in the study area provide an opportunity to recognize the wind effects. Typhoon-induced landslides in Southeast China primarily occurred in the area of planar and convex upward (nose) topography; only 11% of identified cases occurred in areas of concave (hollow) topography (Section 2). Trees in a concave upward topography can be more protected from the wind load (Hales et al., 2009). Hales et al. (2009) and Hales and Miniat (2016) suggest that roots developed in a concave upward topography show a lower cohesion and the concave topography often has high concentrations of water. This provides a possible explanation that shallow landslides in many other places are generally confined to medium and steep slopes of topographic hollows (e.g., Acharya et al., 2016; Dietrich & Dunne, 1978; D'Odorico & Fagherazzi, 2003; Reneau et al., 1990). Therefore, the spatial distribution of identified cases in the study area appears to highlight the possible contribution of wind load to the landslide initiation. Furthermore, the identified typhoon-induced landslides are clustered on slopes covered by trees and bamboo. Trees and bamboos have stronger roots than shrubs and can provide bigger resistance, but their large frontal area will transfer a larger wind load to the slope. Therefore, combined with the preliminary wind tunnel test performed by Sun et al. (2021), who suggest that wind load can work as a driving force to decrease the slope stability, we suggest that wind load is influential to the initiation of typhoon-induced landslides in the study area.

Scenarios designed for the modelling are shown in Table 4. Simulations performed in the absence of rainfall infiltration (Figure 7a) show that F_s of the generalized model decreases from 1.405 (in the initial state) to 1.085 when a Force 14 typhoon (mean wind velocity of 45 m/s) blows downhill. The F_s reduction of 23% indicates the great destabilizing effect of strong wind load. The simulated potential sliding surface, which has the lowest F_s, is located at the rock–soil interface. In this condition, even if the slope is stable (F_s > 1), the slope is in a very dangerous state and a short period of rainfall may cause the slope to fail. Nevertheless, this destabilizing effect seems hard to fail the slope when the typhoon blows uphill. The driving force resulting from the inertia force of tree motion only decreases the F_s by 8% (from 1.405 to 1.293) in a Force 14 typhoon, which is similar to the impact of Force 7 wind that blows downhill, indicating its weak impact. The most dangerous surface (with the lowest factor of safety) is the same as Figure 7b.

Additional simulations were performed on the rainfall contribution with and without preferential infiltration (Figure 7c,d). The rainfall

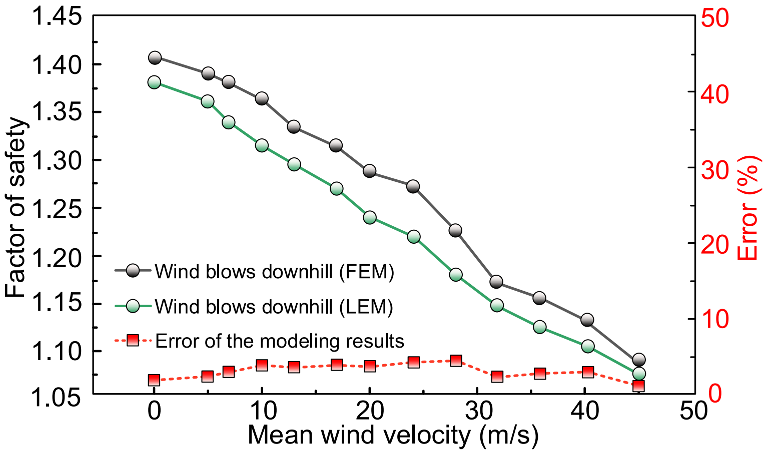

FIGURE 5 Validation of the proposed two-dimensional finite element method (FEM) slope model. The black line shows the FEM modelling results. The green line shows the calculated results using the limited equilibrium method (LEM) method proposed by Zhuang et al. (2022). [Color figure can be viewed at wileyonlinelibrary.com]

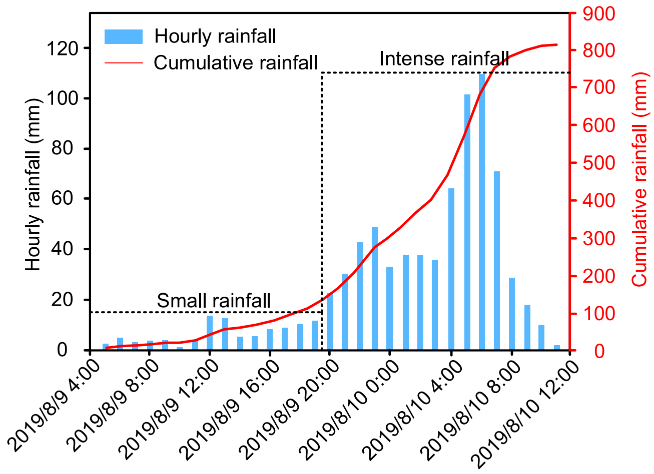

FIGURE 6 Rainfall data of Typhoon Lekima (2019) recorded by Yongjia meteorological bureau in Zhejiang, China. [Color figure can be viewed at wileyonlinelibrary.com]

infiltration over the slope surface shows weak impacts on pore water pressure and slope stability. The simulated F_s shows a value of 1.360 at 12 h (decreased by 3.2%) and remains a value of 1.050 throughout the typhoon-associated precipitation, completely inconsistent with our field investigations (87% of landslides occurred 12 h after the typhoon landing). The upper soil layer with low permeability makes rainwater difficult to rapidly penetrate into the slope. According to Figure 7c, in the case that only considers the uniform infiltration, the rainwater reaches the rock–soil interface at ~22 h and the monitoring point is unsaturated even at the end of the typhoon event. However, the preferential infiltration (in the intense rainfall stage) from the root–soil interface appears to be more influential than the process of slope surface infiltration (uniform infiltration). The rainwater rapidly penetrates within the slope and reaches the rock–soil interface at ~8 h after the rainfall initiation (Figure 7c). The simulated F_s decreases over time to a value less than 1 at 18 h, indicating that the slope becomes unstable. Accounting for preferential flow accelerates the rainfall infiltration within the slope, causing rapid saturation and plastic deformation near the rock–soil interface (Figure 7d).

Using the numerical model, we further explored how wind load and preferential infiltration jointly influenced slope stability (Figure 7e,f). The wind velocity variation of a specific place during a typhoon is extremely complex, depending on the atmosphere and topographic conditions, and is impossible to be accurately predicted (Weisberg & Zheng, 2006). Typhoon measurements performed by Mitsuta and Yoshizumi (1973) show that wind and rainfall intensity showed periodic variations with the approximately same period. The rainfall intensity used in this study shows an increased tendency until it reaches the peak value (Figure 6). Due to the lack of real-time wind velocity data, the wind load (Force 3–14) was assumed to increase linearly in the cases and reached the peak value at the moment of maximum rainfall intensity. The coupled impact of wind load and preferential infiltration has been confirmed to substantially decrease the F_s of the slope. The mode analysis of different wind forces reveals that the simulated hillslope becomes unstable for all the scenarios accounting for the preferential infiltration and wind load (Figure 7e, F_s decreases to less than 1 in scenarios S5–S10), and variations in wind force greatly accelerate the slope failure. A Force 4 wind accelerates the slope failure for ~2 h (at 16 h) relative to the hypothetical scenario where only infiltration is considered (at 18 h, Figure 7c). A Force 14 typhoon and preferential infiltration jointly trigger the landslide just at 10 h following the storm. This phenomenon matches the results in Figure 7a well: A strong wind load might fail the trees-covered slope even with short-time rainfall. The generalized slope failed within ~12 h in cases of a wind force larger than 8. Regarding that a Force 8 wind is common to meet during a typhoon event, the modelling results basically match the actual condition (most cases occurred before and within 12 h after the typhoon landing). Furthermore, two potential sliding surfaces are developed in our generalized model. The landslide develops at the rock–soil interface for the wind with a force of less than 6, showing a similar failure mode to rainfall-induced landslides (Figure 7d). The base of the root–soil composite is in the most dangerous state for a stronger wind load (Figure 7f, shown as an example), presenting consistency with our investigations of partial typhoon-induced landslides that occurred within the upper soil mass (landslides occurred at the weak layer in the overburden soil, as stated in Section 2).

5 | DISCUSSION

5.1 | Initiation of typhoon-induced shallow landslides in Southeast coastal China

Most previous research on typhoon-induced landslides is concentrated on the impact of rainfall infiltration (e.g., Shou & Lin, 2020; Tsai

TABLE 3 Wind velocity used for the numerical simulation (according to Beaufort wind scale).

Wind force	3	4	5	6	7	8	9	10	11	12	13	14
Mean wind velocity (m/s)	5	7	10	13	17	20	24	28	32	36	40	45

TABLE 4 Scenarios modelled in the generalized model for typhoon-induced landslides in Southeast coastal China.

Scenario ID	Applied loading	Slope state
S1	Wind only (blow downhill)	Stable ($F_s = 1.085$)
S2	Wind only (blow uphill)	Stable ($F_s = 1.293$)
S3	Rainfall only (without preferential infiltration)	Stable ($F_s = 1.050$)
S4	Rainfall only (with preferential infiltration)	Unstable (fail at 18.0 h)
S5	Rainfall and Force 4 wind	Unstable (fail at 16.1 h)
S6	Rainfall and Force 6 wind	Unstable (fail at 14.0 h)
S7	Rainfall and Force 8 wind	Unstable (fail at 12.2 h)
S8	Rainfall and Force 10 wind	Unstable (fail at 11.5 h)
S9	Rainfall and Force 12 wind	Unstable (fail at 10.7 h)
S10	Rainfall and Force 14 wind	Unstable (fail at 10.0 h)

et al., 2010). They treated the rainstorm as the primary cause and the rock–soil interface as the potential sliding surface. However, these points can hardly explain why partial typhoon-induced landslides develop within the soil layer (occurred at the weak layer in the overburden soil, as stated in Section 2) and why they occur at approximately the same time as typhoon strikes (even with several metres of landslide depth). Our work provides a likely explanation for these phenomena. Numerical results confirm that the slope surface infiltration (uniform infiltration) is insufficient to account for the typhoon-induced slope failures in Southeast coastal China, yet trees play an important role in destabilizing the slope. Specifically, the driving force caused by the typhoon wind load and preferential infiltration from the root–soil interfaces greatly destabilizes the trees-covered slope.

Using the numerical model, we were able to explore how variations in wind force and wind direction influenced slope stability. Strong wind load appears to have substantial impacts on the slope stability when blowing downhill. In the most dangerous scenario (Force 14 typhoon blows downhill), the calculated F_s reduces by ~23%, still in a stable condition (larger than 1), and the common typhoon-associated rainstorm can easily fail the slope in a short time. In contrast, this detrimental mechanism shows a weak impact when the wind blows in the opposite direction, indicating that the driving force resulting from the tree motion can hardly directly fail the slope. In the absence of rainfall infiltration, the rock–soil interface is in a more dangerous state relative to the base of root–soil composite (Figure 7b). We infer that the higher soil moisture content near the initial underground water table decreases its shear strength and thus forms a landslide-prone region.

Rainfall infiltration over the slope surface has been numerically confirmed to show a weak impact on the slope stability, but its coupled action with preferential infiltration from root–soil interfaces can destabilize the slope at the rock–soil interface in a short time (Figure 7c). For typhoon-induced landslides in Southeast coastal China, it is difficult for the trees-covered slope to fail within hours after typhoon landing if only slope surface infiltration is accounted for. Our results were supported by those of Zhang et al. (2019), who conducted field experiments in the study area (Southeast coastal China) and concluded that slope surface infiltration cannot control the slope stability during a typhoon due to the low permeability of the soil layer. This special hydrological mechanism of preferential infiltration reasonably explains why typhoon-induced landslides have no lag-time effect. Notably, soil permeability depends heavily on the soil properties and is different from place to place. Our conclusion is applicable to the landslide cases in Southeast coastal China and encourages scientists to account for the tree effects when evaluating the slope stability of a typhoon-induced landslide.

Further numerical investigations on the joint impact of wind load and rainfall infiltration demonstrate two failure modes for typhoon-induced landslides. For our generalized model, the slope is prone to fail at the rock–soil interface when the wind with force less than 6 blows in the downhill direction, while the base of root–soil composite is in a more dangerous state with a stronger wind load. Rainwater penetrates through the upper soil as unsaturated infiltration (Li et al., 2016) and always forms a transient saturation zone near the impervious layer (rock–soil interface), resulting in the prior wetting of root–soil region but in an unsaturated state. Roots can provide extra resistance and greatly reinforce the rooting area, and the interface with the lower unreinforced area (base of the rooting area) can be treated as a weak layer. Thus, the slope will destabilize near the maximum depth of the root–soil composite before rainwater saturates the rock–soil interface, when the total driving force (attributed to wind, soil mass and tree mass) exceeds the soil resistance. Additionally, our *in situ* observations also identified landslides developed on the windward slope, meaning the typhoon blows uphill. Given that the driving force results from tree motion show weak impacts on the slope stability (Figure 7a), we suggest that the wind load itself could facilitate the slope failure in the cases but is not the key reason. Typhoon primarily destabilizes the slope through creating extra preferential flow paths (large cracks from root–soil separation) to accelerate the rainfall infiltration. This point is supported by our preliminary wind tunnel test (Sun et al., 2021), which found that violent tree swaying in a typhoon can intensify this preferential hydrological effect and this preferential hydrological effect can increase the regional (near the tree) soil permeability several times than the original state. Therefore, typhoon-induced landslides in Southeast China show different failure mechanisms from those triggered only by rainfall, though rainfall contributed substantially in both scenarios.

While root reinforcement and beneficial hydrological effects of rainfall interception and water transpiration are well established, direct comparisons between the slope stability of forested and unforested slopes are complicated in many natural settings. Roots stabilize the slope through mechanical resistance in tension and friction with soil, causing a limited reinforced region (Conedera & Schwarz, 2018). Our results suggest similar behaviours, with roots only hindering the

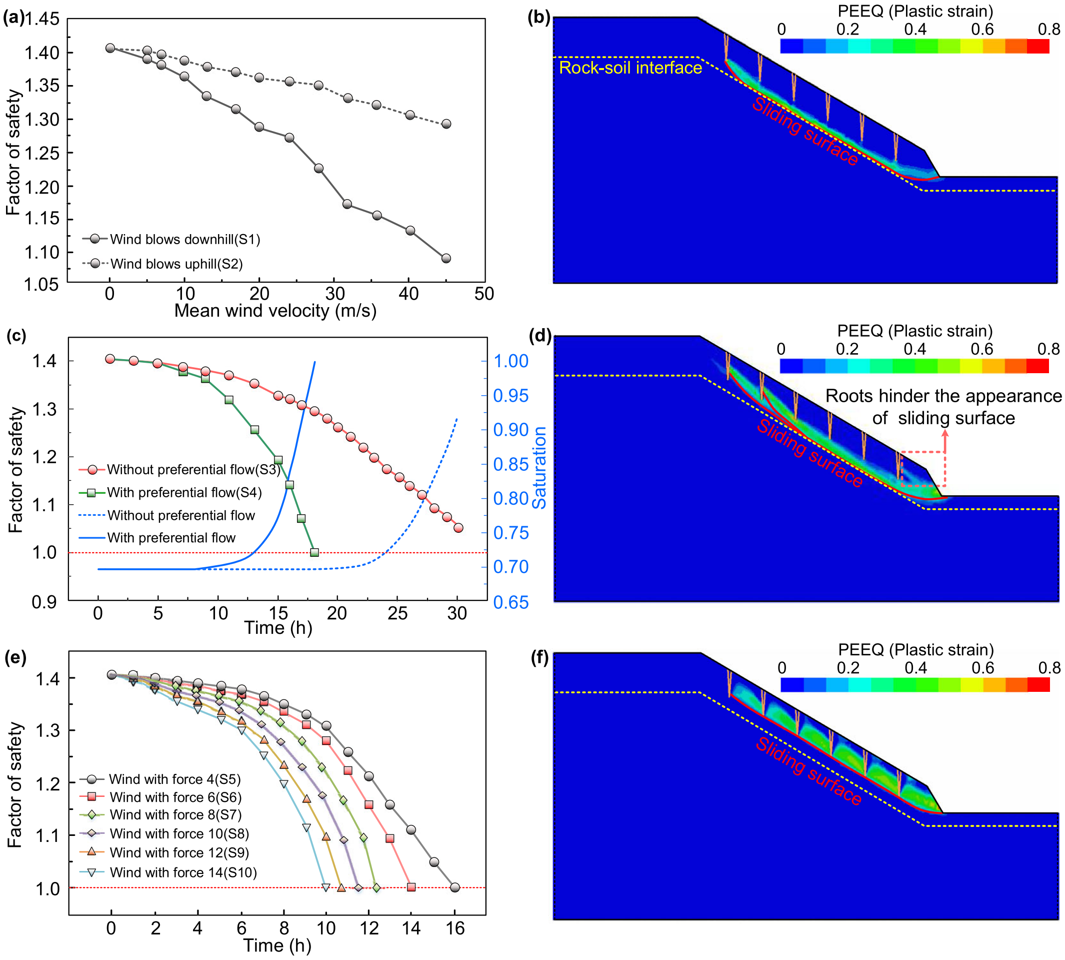

FIGURE 7 Numerical modelling results of the generalized model, indicating the joint impacts of wind load, infiltration process and root reinforcement on the slope stability. (a) Factor of safety with various wind velocities and directions (no wind and wind force of 3–14). (b) Failure mode of the trees-covered slope triggered by wind only. (c) Variation of slope factor of safety and saturation of monitoring point A with and without preferential infiltration. (d) Failure mode of the trees-covered slope triggered by infiltration only. (e) Joint impact of wind load and infiltration on the slope stability. (f) Failure mode of the trees-covered slope triggered by the combination of strong wind load and preferential infiltration (wind with Force 12, shown as an example). [Color figure can be viewed at wileyonlinelibrary.com]

appearance of sliding surface in the root–soil composite region but showing weak impact in the lower layer (Figure 7d). Moreover, beneficial hydrological effects have been experimental and numerically supported to have only minor impacts on the slope stability during intense rainstorms (McGuire et al., 2016). It is thus hard to be optimistic about their stabilization impact in a typhoon. Notably, trees stabilize the slope through root reinforcement, while the appearance of roots can also influence the hydrological process by creating preferential infiltration to destabilize the slope. This mechanism was also supported in the loess region (Guo et al., 2020), where the base of the root–soil composite was identified as the potential sliding surface for most vegetation-covered slopes. The complex interactions between trees and slopes in natural settings necessitate a careful assessment of tree impacts on the slope stability, rather than just considering the beneficial impacts. A key finding of this research is that in Southeast coastal China, the detrimental hydrological processes (preferential infiltration), combined with strong wind load, might ever outweigh the beneficial effects of root reinforcement and moisture reduction in a typhoon event. The coupled impact of strong wind, intense rainfall and trees was responsible for destabilizing soil on the slope, which led to the rapid landslide initiation.

5.2 | Change of trees impacts in different conditions

A question raised by the work is whether our findings can be generalized in typhoon-induced landslides worldwide. Previous research on typhoon (cyclone)-induced landslides in New Zealand shows different results. Marden and Rowan (1994) found landslide rates during Typhoon Bola in 1988 to be 16 times higher on pastures than under pine forested areas. Furthermore, in Wairoa, hundreds of landslides were developed in the deforested grassland areas in April 2022 (Petley, 2022). However, this phenomenon is not entirely contradictory to the findings of our work.

Reid and Page (2002) investigated Typhoon Bola-induced landslides in the Waipaoa catchment. Most shallow landslides are planar failures that involve only residual soil mass and a thin layer of weathered bed-rock, indicating that the sliding surfaces are primarily developed near the soil–bed-rock interface. The soil thickness is less than 2 m, and approximately 55% of the failures are 0.5–0.9 m in depth. Landslides in the area are prominent on slopes with an angle of 25–35° (Dymond et al., 1999). According to measurements of vegetation biomass in New Zealand (Kc et al., 2018; Waston et al., 1999), pine

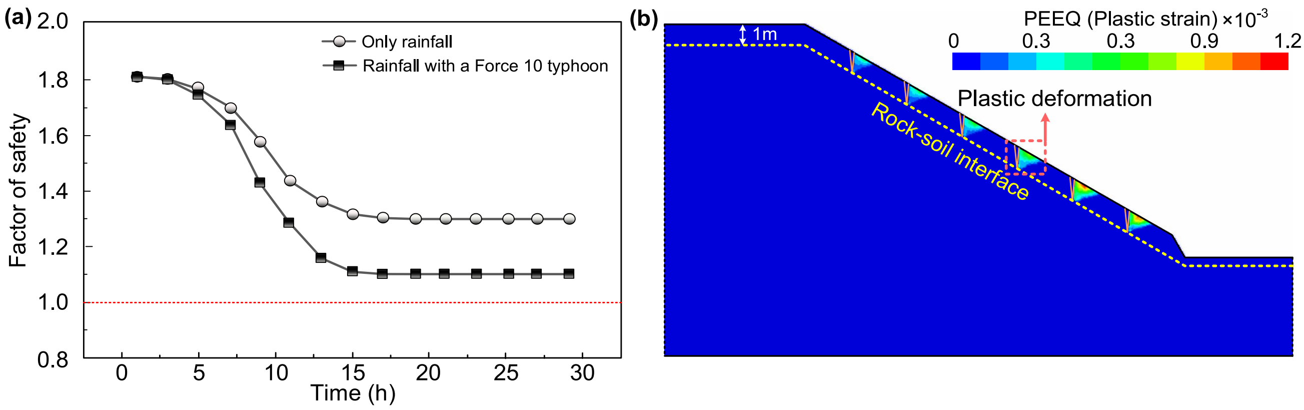

FIGURE 8 Modelling results of a slope with a 1-m-thick soil layer, generalized by cases in New Zealand. (a) Variation of slope stability subjected to rainfall and wind load. (b) Regional plastic deformation developed near the slope surface. [Color figure can be viewed at wileyonlinelibrary.com]

forests (*Pinus radiata*) have a mean maximum root depth of about 1.8 m, while the value for most pasture roots is within 0.3 m. Therefore, we suggest that the forest roots in the area can commonly penetrate through the soil layer and reach the weathered bed-rock. The forests in the area played a role in soil conservation and reinforcement, which greatly stabilized the slope throughout a typhoon. The impact of trees in a typhoon event depends heavily on the meteorological and geological settings. We infer that trees play a stabilizing role in the slopes with a thin soil layer (e.g., Waipaoa). The root system could penetrate into the weathered bed-rock, alleviate the erosion throughout a typhoon and provide considerable resistance to stabilize the soil. For the hillslopes with a relatively thicker soil layer (e.g., Southeast coastal China), trees hinder the slope failures within the root–soil composite, while the preferential flow and strong wind load lead to low F_s for surfaces in the lower region.

Tentative simulations were performed to confirm the above deduction (Figure 8). According to the landslide characteristics in the area, the slope is modelled with a 1-m-thick soil layer, a 30° slope angle, and roots penetrate through the layer. According to measurements performed by Ekanayake and Phillips (1999), the effective frictional angle and cohesion are 16° and 6.8 kPa, respectively. *P. radiata* in the study area has a taproot system but fibrous roots in the shallow layer (Balneaves & De La Mare, 1989; Mason, 1985) and also simplified as piles here. *P. radiata* in the area shows a planting density of 1250–2200 stems/ha (tree distance of about 2.2–3 m) (Marden et al., 2023), and a distance of 3 m is selected for the modelling. Due to the lack of measured permeability coefficient, it is set to be the same with soil mass in Southeast coastal China regarding that they are both residual soils. We suggest that a 2D model is applicable here, because most landslide cases are planar failures and the three-dimensional water concentration/disperse effect is not influential.

Results indicate that the slope remains stable throughout the typhoon, even though the soil layer is fully saturated (Figure 8a). Therefore, even though the permeability coefficient for the modelling is not determined from measurement, its value will not greatly change the final slope stability. Moreover, roots contribute a lot to holding the soil in place and hinder the development of the sliding surface. The conclusion matches the observations in Wairoa in April 2022: The local soil layer could not soak up any more water while the trees-covered slopes were still stable (Petley, 2022). Notably, regional plastic deformation is identified near the slope surface when the trees are subject to a powerful wind load (Figure 8b). This phenomenon is reasonable in a typhoon, but we acknowledge that the plastic zone might be overestimated. The fibrous root system, which is not considered in the present model, could greatly reinforce the region near the land surface in such a scenario.

5.3 | Model limitations and implications

Compared with our preliminary analysis presented in Zhuang et al. (2022), who analysed the initiation of typhoon-induced landslides with an LEM without accounting for the infiltration process, the main contribution of our work is establishing a simple 2D plane strain model to analyse the infiltration process (including preferential infiltration) and its combined effects with wind load and root reinforcement on the slope stability. Furthermore, through comparison analysis with cases in New Zealand, we suggest that trees play different roles in different conditions. However, the simplifications in the model involve the consequences that need to be addressed.

The model is a 2D plane strain mode without considering the water concentration (or disperse) that arises from a three-dimensional topography. Accounting for that typhoon-induced landslides in Southeast coastal China are clustered in the area of planar and convex upward topography (only 11% of cases occurred in areas of concave topography), we suggest that the water-concentrating effect is not very influential. This water-concentrating effect contributes to slope instability and is common in a concave topography because water tends to move towards regions characterized by lower water potential. However, for cases of a convex upward topography, the water dispersive will lead to an increase in slope stability. For shallow landslides with a depth of 1–3 m, the one-dimensional infinite slope model is frequently used for analysis (Cho, 2014). Furthermore, a 2D plane strain slope model is widely used in the modelling of rainfall-induced landslides or root-reinforced slopes and has been confirmed to be a valid approach (e.g., Gao et al., 2021; Liang et al., 2015; Ng et al., 2016). Therefore, we suggest that a 2D plane strain slope model is suitable for the research topic of this paper.

For the wind load calculation, it is assumed to be applied on a single tree without accounting for the impact of a dense forest. We acknowledge that the presence of nearby trees will reduce the wind force (Feistl et al., 2015) and a reduction coefficient is suitable here, but it is extremely hard to determine the specific value of this coefficient because it should be a function of tree type and states. In this study, the wind load is applied on a single tree to simulate the most dangerous scenario. This most dangerous scenario might overestimate the wind effect but is good for disaster mitigation.

Tree roots are modelled as piles insert into the ground. Incorporation of trees into a 2D slope stability analysis is complicated because of their complex mechanical and hydrological effects. Pines (*P. massoniana*) universally distributed in Southeast coastal China have strong taproot systems but sparse lateral roots (Yang et al., 2021; Yao & Wang, 2020; Zhang, Liao, et al., 2022; Zhang, Yang, et al., 2022). Liang et al. (2015) and Wu (2013) conducted numerical work and laboratory tests and showed great performance in simulating taproots roots as piles. Therefore, taproots are modelled as piles in the study; further validations of using Equation (5) check the simplification. Ignoring the effects of fibrous roots would lead to an underestimation of extra resistance provided by the root reinforcement but have minor impacts on the region lower than the rooting zone. We suggest that modelling taproots as piles is an applicable approach in this study.

Our modelling work involves different scale issues, ranging from a small scale (less than a metre) of soil properties and root properties to a large scale (several kilometres) of the typhoon. For the soil properties and root systems, they show a very small scale and change place by place. In this study, median quartile values of measured soil properties are used to ensure the universality of the generalized model. Karthik et al. (2022) performed a sensitivity analysis of slope stability using a 2D plane strain FEM model and indicated that the factor of safety of slopes is sensitive to the soil strength and slope angle. Therefore, accurate parameters arising from measurements will be helpful when conducting stability analysis of a specific slope. As for the large-scale typhoon, the variation of wind velocity and direction is complex and even the primary travel path of a typhoon is hard to predict (Rüttgers et al., 2019). Therefore, we suggest calculating the most dangerous scenario to help disaster mitigation, because it is impossible to accurately predict the velocity and direction of wind at a specific place. Notably, the proposed model and conclusions in this study are applied to forested cut slopes and are not necessarily in all other settings. For unaffected, completely natural, pristine, forested slopes, modelling work could refer to this study to investigate the impacts of trees on slope stability.

Given that present-day slopes in hilly coastal regions are commonly covered by trees, and our results in Southeast China suggest landslide susceptibility is possibly governed by the local detrimental mechanical and hydrological effects of trees during typhoon events. Stability analysis of typhoon-induced landslides in regions with similar geological settings to Southeast China should account for the destabilization impacts of trees rather than treating them as slope failures triggered only by slope surface infiltration. In this study, wind load and preferential infiltration from root–soil gaps are suggested to be influential to slope stability, and root reinforcement is constrained by the limited growth depth of roots. Beneficial hydrological mechanisms of canopy interception and plant water transpiration are known to disappear during a rainstorm (McGuire et al., 2016). This implies that trees play different roles in different conditions, and their contributions are highly related to the geological settings and meteorological conditions. Therefore, further in-depth and targeted work is needed to investigate the impacts of trees on the slope stability in different conditions. Regional-scale analysis of slope stability that aims to predict the spatiotemporal distribution of landslides throughout a typhoon event could be substantially improved by characterizing variations in meteorological conditions, geological settings and vegetation type.

6 | CONCLUSIONS

According to our investigation results, typhoon-induced landslide cases in Southeast China have a landslide depth (depth of the sliding surface) of <3 m and are predominantly at trees-covered slopes of 20–40°. Their spatiotemporal distribution depends heavily on typhoon characteristics, as most landslides are generated within the wind circle and the region of heavy rainfall. In this study, a novel generalized model was proposed to assess the slope stability, and the results were compared with cases in Waipaoa and Wairoa (New Zealand). We found that trees play a destabilizing role in Southeast China but a stabilizing role in the cases in New Zealand. Typhoon-induced landslides in Southeast coastal China are triggered by the joint action of wind, vegetation and rainfall, while only slope surface infiltration (uniform infiltration) is insufficient to account for the slope failure in typhoon events. The detrimental tree impacts of transmitting wind load and preferential infiltration, which were rarely considered in the previous studies, are significantly influential. For the generalized model, a Force 14 typhoon can reduce the F_s by 23%, presenting a significant detrimental contribution. Without accounting for the impact of preferential flow, the simulated F_s is always >1, indicating stable conditions. Our results in Southeast China suggest landslide susceptibility is possibly governed by the detrimental impacts of trees during extreme typhoon events. Further targeted research is needed to specify the meteorological and geological conditions that must be met for the destabilization effects of trees. This is important to the landslide slope stability analysis and disaster mitigation in coastal regions.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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